

Homework 7

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Question 1

Using the mtcars dataset, are cars with automatic or manual transmission more fuel efficient? Support your answer numerically and visually using tidyverse and ggplot tools.

```
# Load necessary libraries  
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.2.3
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tibble' was built under R version 4.2.3
```

```
## Warning: package 'tidyr' was built under R version 4.2.3
```

```
## Warning: package 'readr' was built under R version 4.2.3
```

```
## Warning: package 'purrr' was built under R version 4.2.3
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
## Warning: package 'stringr' was built under R version 4.2.3
```

```
## Warning: package 'forcats' was built under R version 4.2.3
```

```
## Warning: package 'lubridate' was built under R version 4.2.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2   3.5.1      v tibble    3.2.1
```

```
## v lubridate 1.9.3      v tidyr     1.3.1
```

```
## v purrr     1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

# Load mtcars dataset
data(mtcars)

# Convert 'am' variable to a factor (0 = Automatic, 1 = Manual)
mtcars <- mtcars %>%
  mutate(am = factor(am, labels = c("Automatic", "Manual")))

# Summarize fuel efficiency by transmission type
summary_mpg <- mtcars %>%
  group_by(am) %>%
  summarise(
    mean_mpg = mean(mpg, na.rm = TRUE),
    median_mpg = median(mpg, na.rm = TRUE),
    sd_mpg = sd(mpg, na.rm = TRUE),
    count = n()
  )

print(summary_mpg)

```

```

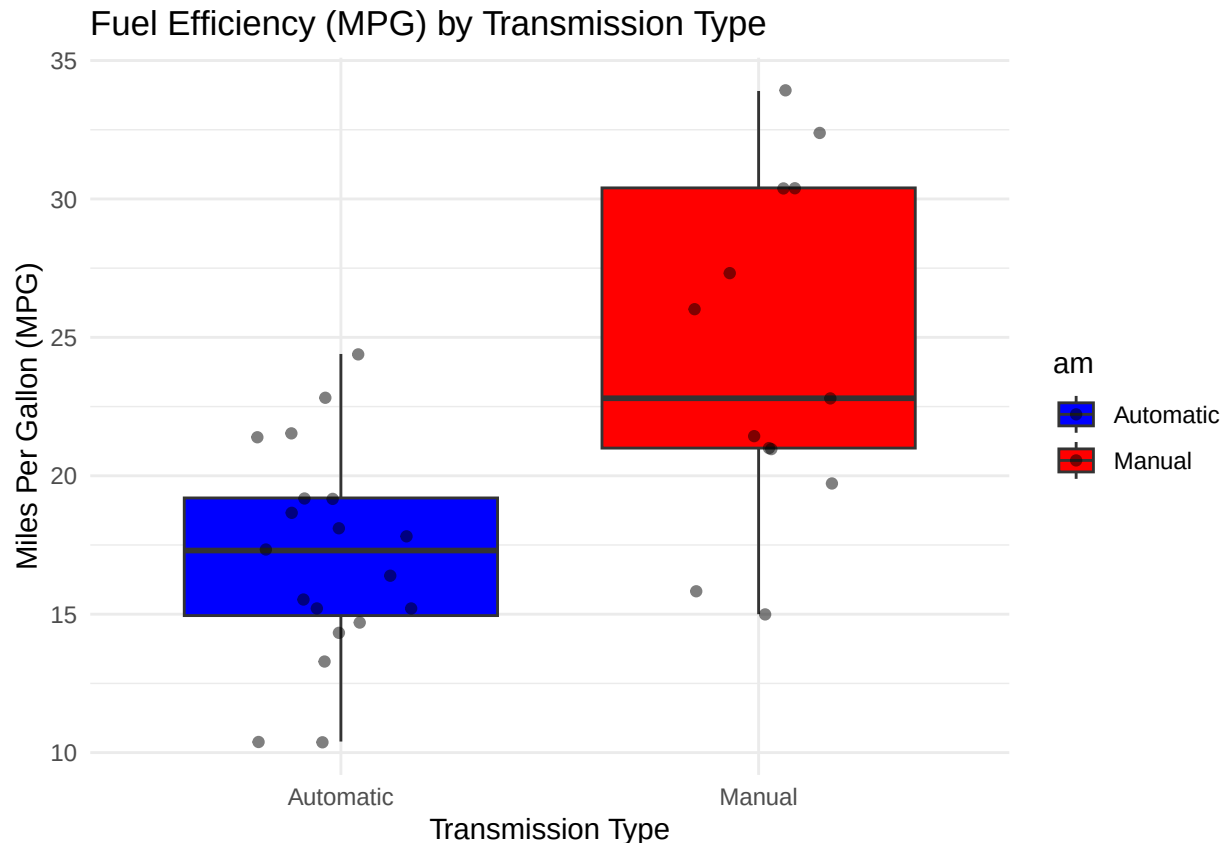
## # A tibble: 2 x 5
##   am      mean_mpg median_mpg sd_mpg count
##   <fct>      <dbl>      <dbl> <dbl> <int>
## 1 Automatic    17.1        17.3   3.83    19
## 2 Manual       24.4        22.8   6.17    13

```

```

# Visualizing the comparison
ggplot(mtcars, aes(x = am, y = mpg, fill = am)) +
  geom_boxplot() +
  geom_jitter(width = 0.2, alpha = 0.5) +
  labs(title = "Fuel Efficiency (MPG) by Transmission Type",
       x = "Transmission Type",
       y = "Miles Per Gallon (MPG)") +
  theme_minimal() +
  scale_fill_manual(values = c("blue", "red"))

```



The visualization indicates that manual transmission cars are more fuel-efficient than automatic transmission cars. The median miles per gallon (MPG) for manual cars is higher, and they exhibit a wider interquartile range (IQR), with several values exceeding 25 MPG. In contrast, automatic cars have a lower median MPG and tend to cluster around the 15-20 MPG range. The data also shows that automatic cars have more values concentrated at the lower end of the MPG spectrum. Overall, the analysis suggests that manual transmission cars offer better fuel efficiency.

Using Numerically and Hypothetically support to the claim.

Null Hypothesis (H₀): There is no significant difference in mean fuel efficiency between automatic and manual transmission cars.

Alternative Hypothesis (H_a): Cars with manual transmission are significantly more fuel efficient compared to automatic transmission cars.

```
# Perform a t-test to check statistical significance
t_test_result <- t.test(mpg ~ am, data = mtcars, alternative = "less")
print(t_test_result)
```

```
##
## Welch Two Sample t-test
##
## data: mpg by am
## t = -3.7671, df = 18.332, p-value = 0.0006868
## alternative hypothesis: true difference in means between group Automatic and group Manual is less than 0
```

```
## 95 percent confidence interval:
##      -Inf -3.913256
## sample estimates:
## mean in group Automatic    mean in group Manual
##           17.14737           24.39231
```

Since the **p-value is less than 0.05**, we have sufficient evidence to reject the null hypothesis. This suggests that there is a statistically significant difference in fuel efficiency between manual and automatic transmission cars. Specifically, cars with manual transmission are significantly more fuel-efficient compared to automatic transmission cars.